DD-W-131c

NOVEMBER 10, 1958

SUPERSEDING
Int. Fed. Spec. DD–W–00131b (HEW–PHS)
November 5, 1957, and
Fed. Spec. DD–W–131a
April 18, 1952

FEDERAL SPECIFICATION

WATCH GLASS, LABORATORY

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.**—Watch glasses covered by this specification are intended for general laboratory uses, such as covering beakers, etc.

1.2 **Classification.**

1.2.1 *Type, classes, and sizes.* — Watch glasses shall be of one type and of the following classes and sizes, as specified.

Class A.—Borosilicate glass.

Class B.—Soda-lime glass.

Approximate diameter (in millimeters)
25
35
50
65
75
90
100
125
137.5
150
175
200

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification.

Federal Specifications:

DD–G–541—Glass (laboratory).

PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.

PPP–B–636—Boxes, Fiber.

PPP–B–676 — Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

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Military Specification:

MIL–L–10547 — Liners, Case, Waterproof.

Military Standard:

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Class A.*—Class A watch glasses shall be of clear borosilicate glass, conforming to the requirements of type I, class A of Federal Specification DD–G–541, and of such quality as to withstand the tests specified in section 4.

3.1.2 *Class B.*—Class B watch glasses shall be of clear soda-lime glass, conforming to the requirements of type II of Federal Specification DD–G–541, and of such quality as to withstand the tests specified in section 4.

3.2 Construction.

3.2.1 *Class A.*—Class A watch glasses shall be well annealed, circular in shape, and concave with uniform fire-polished edges.

3.2.2 *Class B.*—Class B watch glasses shall be well annealed, circular in shape, and concave with finely ground or uniform fire-polished edges, as specified.

3.3 Tolerance. — A tolerance of plus or minus 2 millimeters shall be allowed in size.

3.4 Shock test.

3.4.1 *Class A.*—None of the class A watch glasses shall break or show evidence of crystallization, or clouding upon removal from the autoclave or from the cold immersion bath when tested as specified in 4.3.1.1.

3.4.2 *Class B.*—None of the class B watch glasses shall break when tested as specified in 4.3.1.2.

3.5 Boiling.—None of the watch glasses shall show any evidence of corrosion, scumming, chipping, or cracking when tested as specified in 4.3.2.

3.6 Annealing.—The watch glasses shall show no ring strain and if any longitudinal strain is present, it shall be faint and highly diffused when tested as specified in 4.3.3.

3.7 Workmanship. — The finished watch glasses shall be free from any damage or defects.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, ten specimens or two percent of the delivery (whichever is greater) shall be taken at random from each delivery and tested for compliance with the requirements of this specification.

4.2 Inspection.—The inspector shall ascertain that the watch glasses have been examined and tested in accordance with specification requirements.

4.3 Tests.

4.3.1 *Shock test.*—Watch glasses shall be subjected to the following shock tests as required.

4.3.1.1 *Class A.*—The watch glasses shall be heated in an autoclave at 15 pounds pressure for 15 minutes. They shall then be removed from the autoclave and immediately plunged into water at a temperature of 3° to 7° C. (See 3.4.1)

4.3.1.2 *Class B.*—The watch glasses shall be completely immersed in water and boiled for five (5) minutes, after which they shall be immediately removed and plunged into water

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at 18.5° ± 2½° centigrade (C.). A sufficient volume of water shall be used in the boiling vessel to assure that the watch glasses remain completely immersed during the five (5) minute boiling cycle. (See 3.4.2.)

4.3.2 *Boiling test.* — The watch glasses taken for tests shall be completely immersed in distilled water in a closed vessel having a small steam vent and boiled continuously for 6 hours, boiling distilled water being added when necessary to compensate for evaporation losses. The watch glasses shall be suspended in the water so that they do not come in contact with the containing vessel. (See 3.5.)

4.3.3 *Annealing.*—The watch glasses shall be viewed in a polariscope for compliance with 3.6.

4.4 **Supplier responsibility for inspection and tests.**—Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order.

5. PREPARATION FOR DELIVERY

For civil agency procurement the definitions and applications of the levels of packaging and packing shall be in accordance with Federal Standard No. 102.

5.1 Packaging.

5.1.1 *Packing variation permitted.*—If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.2 *Level A.*

5.1.2.1 *Unit package.*—Each unit shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1, or PPP–B–676. Carton design shall include interior fittings or sufficient cushioning material to insure against scratching or breakage. Each carton shall be adequately secured to prevent accidental opening.

5.1.2.2 *Intermediate package.* — Intermediate package shall be a carton of appropriate size constructed in accordance with Federal Specification PPP–B–636, type I, class 1. Each carton shall be adequately secured to prevent accidental opening.

5.1.3 *Level C.*—Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table IV. The gross weight of the shipping container shall not exceed 200 pounds.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping, when required, shall be in accordance with applicable box specification.

Note.—**Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.**

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5.2.2 *Level B.*

5.2.2.1 *Exterior container.*—Exterior container shall be constructed in accordance with Federal Specification PPP–B–621, table II. The gross weight of the shipping container shall not exceed 150 pounds.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 **Marking.**

5.3.1 *Civil agencies.* — In addition to any special marking required in the contract or order, marking of shipping containers shall be in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*

5.3.2.1 *Interior packages.* — Each unit package and each intermediate package shall be marked in accordance with section 5 of Military Standard MIL–STD–129.

5.3.2.2 *Shipping container.*—Shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipments from the contractor's plant to overseas activities.

6. NOTES

6.1 **Ordering data.**—Purchasers should exercise any options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Class and size required (see 1.2.1).

(*c*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 **Transportation description.**—Transportation description applicable to this item is:

Glassware, not otherwise indexed by name.

Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weight 24,000 pounds, subject to Rule 34, National Motor Freight Classification.

6.3 This specification covers the following items appearing in the Armed Services Medical Stock List:

6640–447–4200 WATCH GLASS, LABORATORY, 25 mm., 12s.

6640–447–4400 WATCH GLASS, LABORATORY, 50 mm., 12s.

6640–447–4600 WATCH GLASS, LABORATORY, 137.5 mm., 12s.

6640–273–6960 WATCH GLASS, LABORATORY, 35 mm., 12s.

6640–273–6962 WATCH GLASS, LABORATORY, 75 mm., 12s.

6640–273–6963 WATCH GLASS, LABORATORY, 100 mm., 12s.

6.3.1 *Packaging quantities.*—The number of units indicated shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in procurement document.

6.4 Class A, Borosilicate glass and Class B, Soda-lime glass were designated as "Class A, Resistance glass" and "Class B, Ordinary glass", respectively in Federal Specification DD–W–131a.

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Packaging quantities

Federal Stock Number	No. of watch glasses constituting one unit	Unit package	Intermediate package	Shipping container
6640–447–4200	12	1	24	864
6640–447–4400	12	1	12	288
6640–447–4600	12	1	Not required	48

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MILITARY INTERESTS:

Army—C M
Navy—Md
Air Force.

Copies of this specification may be purchased for 5 cents each.

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